

i have my mcq and programmin exam tomorrow so i want html file related to it like coding questions and in hint ans in java so basically style will be tcs and the exam is Programming and Numeric aability in programming Array or string related stuff and in Numeric ability Percentage , Profit Loss ,Ratio proportion and Number system and Data interpretation ive attached files now make minimal design light theme html files ensure that coding will be in my vs code so just questions and 3 hints+ show code and in Aptitude section wise 25 MCQS will be there per topic and each kinda questions should be covered... after end of it count marks Uploaded files: - 1783101609513_Numerical Ability by Placement Lelo.pdf (2.6 MB) - 1783101609514_TCS-Exam.pdf

I

Python script to build and verify Percentage questions

gen_profitloss.py

```
import json
```

```
Q = []
```

```
def add(text, options, correct_idx, explain):
```

```
    assert len(options) == 4
```

```
    Q.append({"q": text, "options": options, "answer": correct_idx, "exp": explain})
```

```
# 1
```

```
add("A man buys an article for ₹450 and sells it for ₹540. Find his profit percentage.",  
    ["20%", "18%", "22%", "15%"], 0,  
    "Profit = 540-450 = 90. Profit% = 90/450×100 = 20%.")
```

```
# 2
```

```
add("A shopkeeper sells an article at a loss of 12%. If he had sold it for ₹99 more, he  
would have gained 10%. Find the cost price.",  
    ["₹450", "₹500", "₹420", "₹480"], 0,  
    "Let CP = x. 1.10x - 0.88x = 99 → 0.22x = 99 → x = 450.")
```

```
# 3
```

```
add("The cost price of 20 articles is equal to the selling price of 16 articles. Find the profit  
percentage.",  
    ["25%", "20%", "22%", "16%"], 0,  
    "CP of 1 article=x, so CP of 20 = 20x = SP of 16 → SP per article = 20x/16 = 1.25x. Profit%  
= 25%.")
```

4

add("A trader marks his goods 40% above cost price and allows a discount of 25%. Find his profit percentage.",

["5%", "10%", "8%", "15%"], 0,

"CP=100, MP=140, SP=140×0.75=105. Profit% = 5%.")

5

add("By selling a table for ₹1140, a man loses 5%. At what price should he sell it to gain 8%?",

["₹1296", "₹1300", "₹1250", "₹1280"], 0,

"CP = 1140/0.95 = 1200. SP for 8% gain = 1200×1.08 = 1296.")

6

add("A dishonest dealer professes to sell his goods at cost price but uses a weight of 800 gm instead of 1 kg. Find his profit percentage.",

["25%", "20%", "22%", "18%"], 0,

"He gives 800g but charges for 1000g. Profit% = (200/800)×100 = 25%.")

7

add("A man sells two articles, each for ₹1200. On one he gains 20% and on the other he loses 20%. Find his overall profit or loss percentage.",

["4% loss", "No profit no loss", "4% profit", "2% loss"], 0,

"When SP is equal and gain%=loss%, there is always an overall loss of (common%/10)^2 = (20/10)^2 = 4%.")

8

add("A shopkeeper allows a discount of 10% on the marked price of an article and still makes a profit of 26%. If the article costs him ₹450, find the marked price.",

["₹630", "₹600", "₹650", "₹610"], 0,

"SP = 450×1.26 = 567. MP × 0.90 = 567 → MP = 567/0.9 = 630.")

9

add("A man buys a watch for ₹800 and sells it at a loss of 15%. He then buys it back at a price 10% less than what he sold it for, and sells it again at a profit of 20%. Find his overall profit or loss.",

["₹34.80 profit", "₹40 profit", "₹30 loss", "₹36 profit"], 0,

"SP1 = 800×0.85 = 680. Buys back at 680×0.90 = 612. Sells at 612×1.20 = 734.40. Net: spent 800, got 680, spent 612, got 734.40. Net cash flow = -800+680-612+734.40 = 2.40? Let's recompute: overall profit = (680-800)+(734.40-612) = -120+122.40 = 2.40. This needs revision.")

fix 9 - recompute properly and simplify problem to avoid ambiguity

Q.pop()

add("A man buys a watch for ₹800 and sells it at a loss of 15%. He later buys an identical watch for ₹680 and sells it at a profit of 20%. Find his overall profit or loss on the two transactions combined.",

["₹16 profit", "₹20 loss", "₹10 profit", "No profit no loss"], 0,

"Transaction 1: CP=800, SP=800×0.85=680, loss=120. Transaction 2: CP=680, SP=680×1.20=816, profit=136. Net = 136-120 = ₹16 profit.")

10

add("A retailer buys 30 pens at the marked price of 25 pens from a wholesaler. If he sells these pens giving a discount of 5%, what is his profit percentage?",

["14%", "12%", "16%", "10%"], 0,

"Let marked price of 1 pen = ₹1. CP of 30 pens = price of 25 = ₹25. SP of 30 pens at 5% discount = 30×0.95 = ₹28.5. Profit% = (28.5-25)/25×100 = 14%.")

11

add("A man sold an article at a loss of 15%. Had he sold it for ₹300 more, he would have gained 5%. Find the selling price of the article to gain 10%.")

Input validation errors occurred: description: Field required

Input validation errors occurred: description: Field required

Complete the profit-loss question generator with remaining questions 11-25

Successfully replaced string in /home/claude/gen_profitloss.py

Good, that's clean now (25 questions, question 9 properly fixed via pop/re-add).
Let me run it.

Now Ratio & Proportion (25 questions):

Python script to build Ratio and Proportion questions

Now Number System (25 questions):

Python script to build and verify Number System questions

Python script to build Data Interpretation question sets with computed answers

Python script to build coding questions dataset (Java hints+solutions)

Now I'll build the full exam HTML file. Given the scale (125 MCQs + 16 coding problems), I'll assemble it via a Python script for reliability.

build_html.py

```
import json
```

```
perc = json.load(open('percentage.json'))
pl = json.load(open('profitloss.json'))
ratio = json.load(open('ratio.json'))
ns = json.load(open('numsys.json'))
di = json.load(open('di.json'))
coding = json.load(open('coding.json'))
```

```
DATA = {
    "aptitude": {
        "percentage": {"label": "Percentage", "questions": perc},
        "profitloss": {"label": "Profit & Loss", "questions": pl},
        "ratio": {"label": "Ratio & Proportion", "questions": ratio},
        "numsys": {"label": "Number System", "questions": ns},
        "di": {"label": "Data Interpretation", "sets": di},
    },
    "coding": coding
}
```

```
data_json = json.dumps(DATA, ensure_ascii=False)
data_json = data_json.replace("</script", "<\\script").replace("<script", "<\\script")
```

```
print("Data JSON length:", len(data_json))
```

```
with open("data_blob.js", "w") as f:
    f.write("const EXAM_DATA = " + data_json + ";\n")
```

```
print("done")
```

style.css

```
:root{
    --bg:#f3f5f8;
    --panel:#ffffff;
    --border:#e2e6ec;
    --text:#1f2733;
    --muted:#6b7583;
    --blue:#0b5fce;
    --blue-dark:#0847a3;
    --blue-tint:#eaf1fd;
```

```

--green:#1a9c5a;
--green-tint:#e7f7ee;
--red:#d5382f;
--red-tint:#fdecea;
--amber:#b8860b;
--grey-tint:#eef0f3;
--radius:8px;
--mono: "SFMono-Regular", Consolas, "Liberation Mono", Menlo, monospace;
--sans: -apple-system, BlinkMacSystemFont, "Segoe UI", Roboto, Helvetica, Arial, sans-serif;
}
*{box-sizing:border-box;}
html,body{height:100%;}
body{
  margin:0;
  font-family:var(--sans);
  background:var(--bg);
  color:var(--text);
  font-size:14px;
  line-height:1.5;
}
button{font-family:inherit;cursor:pointer;}
::selection{background:#bcd6fb;}

/* ----- Top bar ----- */
.topbar{
  position:sticky;top:0;z-index:50;
  display:flex;align-items:center;gap:16px;
  background:var(--panel);
  border-bottom:1px solid var(--border);
  padding:10px 20px;
}
.brand{display:flex;align-items:center;gap:10px;}
.brand-mark{
  width:30px;height:30px;border-radius:7px;
  background:linear-gradient(135deg,var(--blue),#3b82f6);
  display:flex;align-items:center;justify-content:center;
  color:#fff;font-weight:700;font-size:13px;
}
.brand-text{display:flex;flex-direction:column;line-height:1.15;}
.brand-text b{font-size:14.5px;letter-spacing:.2px;}
.brand-text span{font-size:11px;color:var(--muted);}

```

```

.section-tabs{display:flex;gap:4px;margin-left:8px;}
.section-tab{
  padding:8px 16px;border-radius:7px;border:1px solid transparent;
  background:transparent;color:var(--muted);font-weight:600;font-size:13px;
}
.section-tab.active{background:var(--blue-tint);color:var(--blue-dark);}
.section-tab:hover:not(.active){background:var(--grey-tint);}

.topbar-spacer{flex:1;}

.timer{
  display:flex;align-items:center;gap:6px;
  font-variant-numeric:tabular-nums;
  font-weight:600;color:var(--text);
  background:var(--grey-tint);
  padding:6px 12px;border-radius:20px;font-size:13px;
}
.timer svg{width:14px;height:14px;color:var(--muted);}

.btn{
  border:1px solid var(--border);background:var(--panel);color:var(--text);
  padding:8px 14px;border-radius:7px;font-weight:600;font-size:13px;
}
.btn:hover{background:var(--grey-tint);}
.btn-primary{background:var(--blue);border-color:var(--blue);color:#fff;}
.btn-primary:hover{background:var(--blue-dark);}
.btn-danger{background:#fff;border-color:var(--red);color:var(--red);}
.btn-danger:hover{background:var(--red-tint);}
.btn-ghost{border-color:transparent;background:transparent;}
.btn-ghost:hover{background:var(--grey-tint);}
.btn.disabled{opacity:.45;cursor:not-allowed;}
.btn-sm{padding:5px 10px;font-size:12px;border-radius:6px;}

/* ----- Layout ----- */
.app{display:none;max-width:1300px;margin:0 auto;padding:18px 20px 60px;}
.app.active{display:block;}

.subtabs{display:flex;flex-wrap:wrap;gap:6px;margin-bottom:16px;}
.subtab{
  padding:7px 14px;border-radius:20px;border:1px solid var(--border);
  background:var(--panel);color:var(--muted);font-weight:600;font-size:12.5px;
}

```

```

.subtab.active{background:var(--blue);border-color:var(--blue);color:#fff;}
.subtab .count{opacity:.75;font-weight:500;margin-left:4px;}

.layout{display:grid;grid-template-columns:1fr 300px;gap:18px;align-items:start;}
@media (max-width:880px){.layout{grid-template-columns:1fr;}}

.panel{
  background:var(--panel);border:1px solid var(--border);border-radius:var(--radius);
  padding:22px 24px;
}

/* ----- Question ----- */
.q-meta{display:flex;align-items:center;justify-content:space-between;margin-
bottom:14px;}
.q-tag{
  font-size:11.5px;font-weight:700;color:var(--blue-dark);background:var(--blue-tint);
  padding:3px 9px;border-radius:5px;letter-spacing:.3px;text-transform:uppercase;
}
.q-index{color:var(--muted);font-size:12.5px;font-weight:600;}
.q-text{font-size:15.5px;font-weight:600;margin:0 0 18px;color:var(--text);line-
height:1.55;}

.options{display:flex;flex-direction:column;gap:9px;margin-bottom:22px;}
.option{
  display:flex;align-items:flex-start;gap:11px;
  border:1px solid var(--border);border-radius:7px;padding:11px 14px;
  background:#fff;
}
.option:hover{border-color:#c4d5f2;background:#fafcff;}
.option.selected{border-color:var(--blue);background:var(--blue-tint);}
.option input{margin-top:3px;accent-color:var(--blue);}
.option label{flex:1;cursor:pointer;font-size:14px;}
.option .opt-letter{
  width:20px;height:20px;border-radius:50%;background:var(--grey-tint);
  display:flex;align-items:center;justify-content:center;font-size:11px;font-weight:700;
  color:var(--muted);flex-shrink:0;
}
.option.selected .opt-letter{background:var(--blue);color:#fff;}

/* review states after submit */
.option.correct-ans{border-color:var(--green);background:var(--green-tint);}
.option.correct-ans .opt-letter{background:var(--green);color:#fff;}

```

```

.option.wrong-ans{border-color:var(--red);background:var(--red-tint);}
.option.wrong-ans .opt-letter{background:var(--red);color:#fff;}
.explain{
  margin-top:14px;padding:12px 14px;background:var(--grey-tint);border-radius:7px;
  font-size:13px;color:var(--text);display:none;
}
.explain.show{display:block;}
.explain b{color:var(--blue-dark);}

.q-nav-row{display:flex;justify-content:space-between;align-
items:center;gap:10px;padding-top:6px;border-top:1px solid var(--border);margin-
top:4px;}
.q-nav-row .left-btns,.q-nav-row .right-btns{display:flex;gap:8px;}

/* ----- Palette sidebar ----- */
.side-title{font-size:12px;font-weight:700;color:var(--muted);text-
transform:uppercase;letter-spacing:.4px;margin:0 0 10px;}
.legend{display:flex;flex-wrap:wrap;gap:10px;margin-bottom:14px;font-
size:11.5px;color:var(--muted);}
.legend span{display:flex;align-items:center;gap:5px;}
.dot{width:9px;height:9px;border-radius:50%;display:inline-block;}
.dot.answered{background:var(--green);}
.dot.notanswered{background:var(--red);}
.dot.notvisited{background:#c7ccd4;}
.dot.current{background:var(--blue);}

.palette{display:grid;grid-template-columns:repeat(5,1fr);gap:7px;margin-
bottom:16px;}
.pnum{
  aspect-ratio:1;border-radius:6px;border:1px solid var(--border);
  background:#fff;font-weight:700;font-size:12.5px;color:var(--muted);
  display:flex;align-items:center;justify-content:center;
}
.pnum.answered{background:var(--green);border-color:var(--green);color:#fff;}
.pnum.notanswered{background:var(--red-tint);border-color:var(--red);color:var(--
red);}
.pnum.current{outline:2px solid var(--blue);outline-offset:1px;}
.pnum:hover{border-color:var(--blue);}

.topic-summary{border-top:1px solid var(--border);padding-top:14px;margin-
top:4px;font-size:12.5px;color:var(--muted);}
.topic-summary div{display:flex;justify-content:space-between;padding:3px 0;}

```

```
.topic-summary b{color:var(--text);}
```

```
/* ----- Data interpretation chart ----- */
```

```
.di-chart-box{  
  border:1px solid var(--border);border-radius:8px;padding:18px 20px;margin-  
bottom:18px;background:#fbfcfe;  
}  
.di-chart-title{font-weight:700;font-size:13.5px;margin-bottom:14px;color:var(--text);}  
.bar-chart{display:flex;align-items:flex-end;gap:18px;height:180px;padding:0  
6px;border-bottom:1px solid var(--border);}  
.bar-group{display:flex;flex-direction:column;align-items:center;justify-content:flex-  
end;flex:1;height:100%;gap:3px;}  
.bar-set{display:flex;gap:3px;align-items:flex-end;height:100%;}  
.bar{width:16px;border-radius:3px 3px 0 0;position:relative;}  
.bar .bar-val{position:absolute;top:-16px;left:50%;transform:translateX(-50%);font-  
size:10px;font-weight:700;color:var(--muted);white-space:nowrap;}  
.bar-label{font-size:11px;color:var(--muted);margin-top:8px;font-weight:600;}  
.chart-legend{display:flex;gap:16px;margin-top:12px;font-size:11.5px;color:var(--  
muted);}  
.chart-legend span{display:flex;align-items:center;gap:5px;}  
.legend-swatch{width:10px;height:10px;border-radius:3px;}
```

```
.pie-wrap{display:flex;gap:26px;align-items:center;flex-wrap:wrap;}  
.pie{width:170px;height:170px;border-radius:50%;flex-shrink:0;}  
.pie-legend{display:flex;flex-direction:column;gap:7px;font-size:12.5px;}  
.pie-legend .row{display:flex;align-items:center;gap:8px;}
```

```
.di-table{width:100%;border-collapse:collapse;font-size:12.5px;}  
.di-table th,.di-table td{border:1px solid var(--border);padding:7px 10px;text-  
align:center;}  
.di-table th{background:var(--grey-tint);font-weight:700;color:var(--text);}  
.di-table td:first-child,.di-table th:first-child{text-align:left;font-weight:600;}
```

```
/* ----- Coding section ----- */
```

```
.code-layout{display:grid;grid-template-columns:280px 1fr;gap:18px;align-items:start;}  
@media (max-width:880px){.code-layout{grid-template-columns:1fr;}}  
.code-list{display:flex;flex-direction:column;gap:6px;}  
.code-item{  
  display:flex;align-items:center;gap:10px;padding:10px 12px;border-radius:7px;  
  border:1px solid var(--border);background:#fff;text-align:left;  
}  
.code-item:hover{border-color:#c4d5f2;}
```

```

.code-item.active{border-color:var(--blue);background:var(--blue-tint);}
.code-item .ci-num{
  width:22px;height:22px;border-radius:50%;background:var(--grey-tint);color:var(--
muted);
  display:flex;align-items:center;justify-content:center;font-size:11px;font-
weight:700;flex-shrink:0;
}
.code-item.solved .ci-num{background:var(--green);color:#fff;}
.code-item.hints .ci-num{background:var(--amber);color:#fff;}
.code-item.viewed .ci-num{background:#c7ccd4;color:#fff;}
.ci-body{flex:1;min-width:0;}
.ci-title{font-size:12.5px;font-weight:600;color:var(--text);white-
space:nowrap;overflow:hidden;text-overflow:ellipsis;}
.diff-badge{font-size:10px;font-weight:700;padding:2px 7px;border-radius:10px;text-
transform:uppercase;letter-spacing:.3px;}
.diff-Easy{background:var(--green-tint);color:var(--green);}
.diff-Medium{background:#fff4e0;color:#b8860b;}
.diff-Hard{background:var(--red-tint);color:var(--red);}

.code-title-row{display:flex;align-items:center;gap:10px;margin-bottom:8px;}
.code-title-row h2{font-size:17px;margin:0;}

.io-block{
  background:#0f1720;color:#dbe4ee;border-radius:7px;padding:14px 16px;
  font-family:var(--mono);font-size:12.5px;white-space:pre-wrap;margin:14px 0
18px;line-height:1.6;
}
.io-block .lbl{color:#7fb2ff;font-weight:700;}

.hints-block{display:flex;flex-direction:column;gap:10px;margin-bottom:18px;}
.hint-btn-row{display:flex;gap:8px;flex-wrap:wrap;}
.hint-card{
  display:none;background:var(--blue-tint);border:1px solid #cfe0fa;border-radius:7px;
  padding:11px 14px;font-size:13px;color:var(--text);
}
.hint-card.show{display:block;}
.hint-card b{color:var(--blue-dark);}

.solution-box{margin-top:8px;}
.code-block{
  background:#0f1720;color:#e3eaf2;border-radius:7px;padding:16px 18px;
  font-family:var(--mono);font-size:12.5px;line-height:1.65;white-space:pre;overflow-

```

```

x:auto;display:none;
}
.code-block.show{display:block;}

.status-row{display:flex;gap:8px;flex-wrap:wrap;margin-top:18px;padding-
top:16px;border-top:1px solid var(--border);}
.status-btn{padding:8px 14px;border-radius:7px;border:1px solid var(--
border);background:#fff;font-size:12.5px;font-weight:600;color:var(--muted);}
.status-btn.sel-solved.active{background:var(--green);border-color:var(--
green);color:#fff;}
.status-btn.sel-hints.active{background:var(--amber);border-color:var(--
amber);color:#fff;}
.status-btn.sel-viewed.active{background:#8a92a0;border-color:#8a92a0;color:#fff;}

.code-progress{font-size:12.5px;color:var(--muted);margin-bottom:12px;}

/* ----- Results screen ----- */
.results-overlay{
  position:fixed;inset:0;background:rgba(15,20,28,.55);z-index:200;
  display:none;align-items:flex-start;justify-content:center;overflow:auto;padding:40px
16px;
}
.results-overlay.show{display:flex;}
.results-card{background:#fff;border-radius:12px;max-
width:760px;width:100%;padding:32px 34px 30px;}
.results-card h2{margin:0 0 4px;font-size:20px;}
.results-card .sub{color:var(--muted);font-size:13px;margin-bottom:22px;}
.score-hero{
  display:flex;align-items:center;gap:22px;background:var(--blue-tint);border-
radius:10px;
  padding:20px 24px;margin-bottom:22px;
}
.score-big{font-size:34px;font-weight:800;color:var(--blue-dark);}
.score-big span{font-size:16px;font-weight:600;color:var(--muted);}
.score-pct{font-size:13px;color:var(--muted);}
.results-table{width:100%;border-collapse:collapse;margin-bottom:18px;font-
size:13px;}
.results-table th,.results-table td{padding:9px 10px;border-bottom:1px solid var(--
border);text-align:left;}
.results-table th{color:var(--muted);font-weight:700;text-transform:uppercase;font-
size:10.5px;letter-spacing:.3px;}
.results-table td.num{text-align:center;}

```

```
.results-actions{display:flex;gap:10px;justify-content:flex-end;margin-top:10px;}
```

```
.badge-note{font-size:11.5px;color:var(--muted);background:var(--grey-tint);padding:10px 12px;border-radius:7px;margin-bottom:16px;}
```

```
@media(max-width:600px){  
  .topbar{flex-wrap:wrap;}  
  .section-tabs{order:3;width:100%;}  
  .palette{grid-template-columns:repeat(5,1fr);}  
  .results-card{padding:22px 18px;}  
}
```

Main application JavaScript for the exam interface

shell.html

```
<!DOCTYPE html>  
<html lang="en">  
<head>  
<meta charset="UTF-8">  
<meta name="viewport" content="width=device-width, initial-scale=1.0">  
<title>TCS Mock Test — Programming & Numerical Ability</title>  
<style>  
/* __CSS__ */  
</style>  
</head>  
<body>  
  
<div class="topbar">  
  <div class="brand">  
    <div class="brand-mark">TX</div>  
    <div class="brand-text">  
      <b>TCS-Style Mock Test</b>  
      <span>Programming & Numerical Ability</span>  
    </div>  
  </div>  
  <div class="section-tabs">  
    <button class="section-tab active" data-section="aptitude">Aptitude (125  
MCQs)</button>  
    <button class="section-tab" data-section="coding">Programming (16  
problems)</button>  
  </div>  
  <div class="topbar-spacer"></div>  
  <div class="timer">
```

```
<svg viewBox="0 0 24 24" fill="none" stroke="currentColor" stroke-width="2"><circle
cx="12" cy="12" r="9"/><path d="M12 7v5l3 3"/></svg>
<span id="timerText">00:00:00</span>
</div>
<button class="btn btn-primary" id="btnSubmit">Submit Exam</button>
</div>
```

```
<!-- ===== APTITUDE ===== -->
```

```
<div class="app active" id="appAptitude">
  <div class="subtabs" id="aptSubtabs"></div>
  <div class="layout">
    <div class="panel" id="questionPanel"></div>
    <div class="panel">
      <div class="side-title">Question Palette</div>
      <div class="legend">
        <span><i class="dot answered"></i>Answered</span>
        <span><i class="dot notanswered"></i>Visited</span>
        <span><i class="dot notvisited"></i>Not visited</span>
      </div>
      <div class="palette" id="palette"></div>
      <div class="side-title">This Topic</div>
      <div class="topic-summary" id="topicSummary"></div>
    </div>
  </div>
</div>
```

```
<!-- ===== CODING ===== -->
```

```
<div class="app" id="appCoding">
  <div class="code-progress" id="codeProgress"></div>
  <div class="code-layout">
    <div class="panel">
      <div class="side-title" style="margin-bottom:12px;">Problems</div>
      <div class="code-list" id="codeList"></div>
    </div>
    <div class="panel" id="codePanel"></div>
  </div>
</div>
```

```
<!-- ===== RESULTS OVERLAY ===== -->
```

```
<div class="results-overlay" id="resultsOverlay">
  <div class="results-card">
    <h2>Your Score</h2>
  </div>
</div>
```

```
<div class="sub">Here's how you did across the mock test.</div>  
<div id="resultsBody"></div>  
</div>  
</div>
```

```
<script>  
/*__DATA__*/  
</script>  
<script>  
/*__APP__*/  
</script>  
</body>  
</html>
```